SET 16: CUSTOMER DEMOGRAPHICS ANALYSIS

Dataset	Customer Demographics
Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in $)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

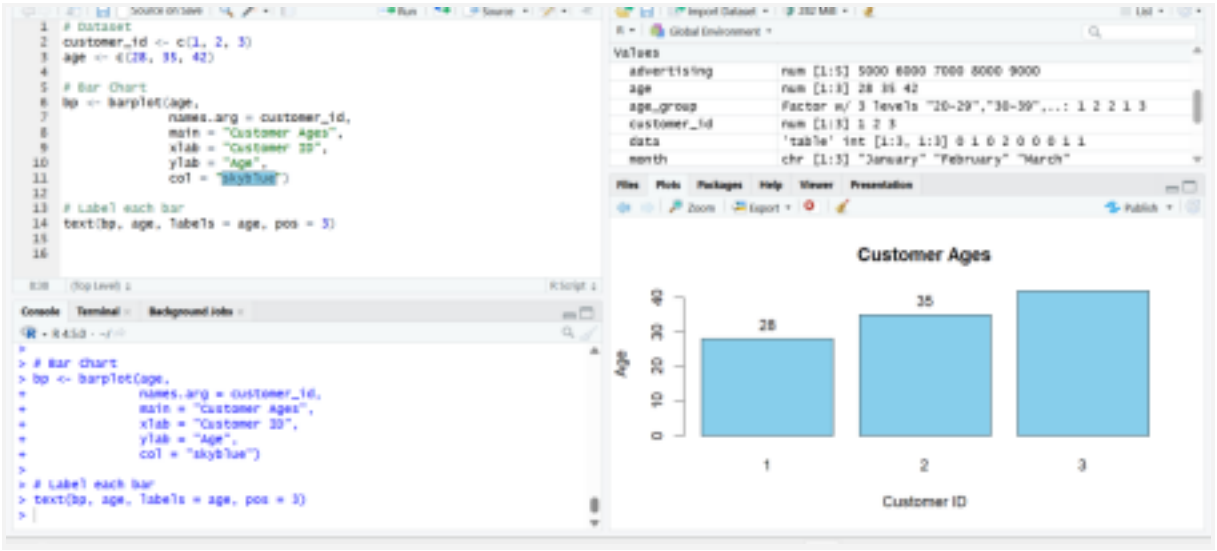
PROGRAM 1: Bar Chart of Customer Ages

Question: Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
customer_id <- c(1, 2, 3)
age <- c(28, 35, 42)

# Bar Chart
bp <- barplot(age,
 names.arg = customer_id,
 main = "Customer Ages",
 xlab = "Customer ID",
 ylab = "Age",
 col = "skyblue")

# Label each bar
text(bp, age, labels = age, pos = 3)
```



Output 1: Customer Ages Bar Chart

Observation: The bar chart represents the ages of the three customers. The ages are 28, 35, and 42 years for Customer IDs 1, 2, and 3 respectively.

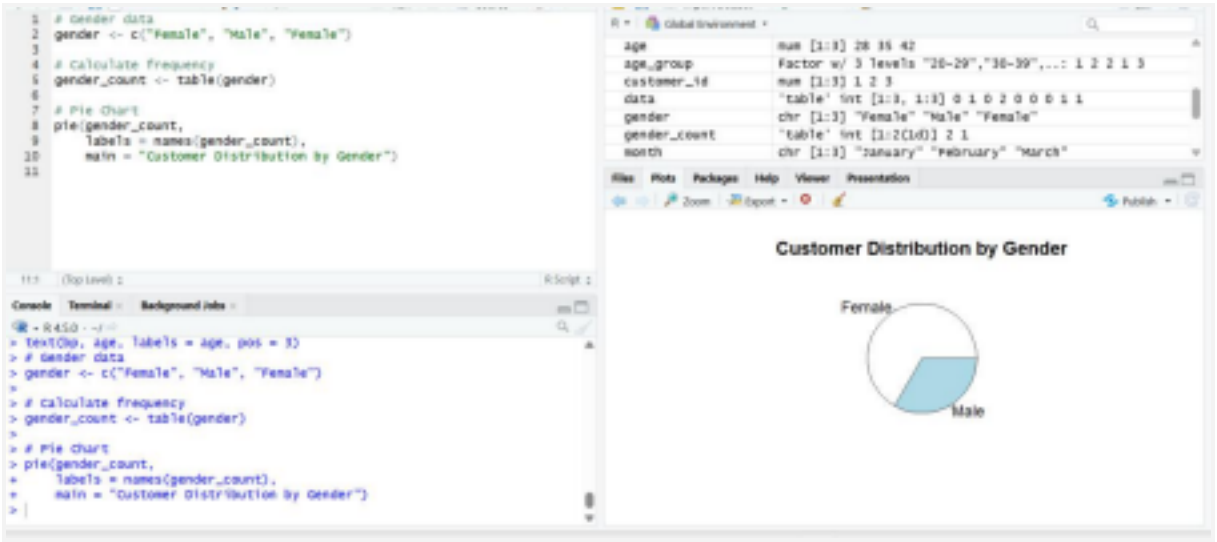
PROGRAM 2: Pie Chart of Customer Gender

Question: Generate a pie chart to represent the distribution of customers by gender.

```
# Gender data
gender <- c("Female", "Male", "Female")

# Calculate frequency
gender_count <- table(gender)

# Pie Chart
pie(gender_count,
 labels = names(gender_count),
 main = "Customer Distribution by Gender")
```



Output 2: Customer Distribution by Gender Pie Chart

Observation: The pie chart shows that there are 2 female customers and 1 male customer in

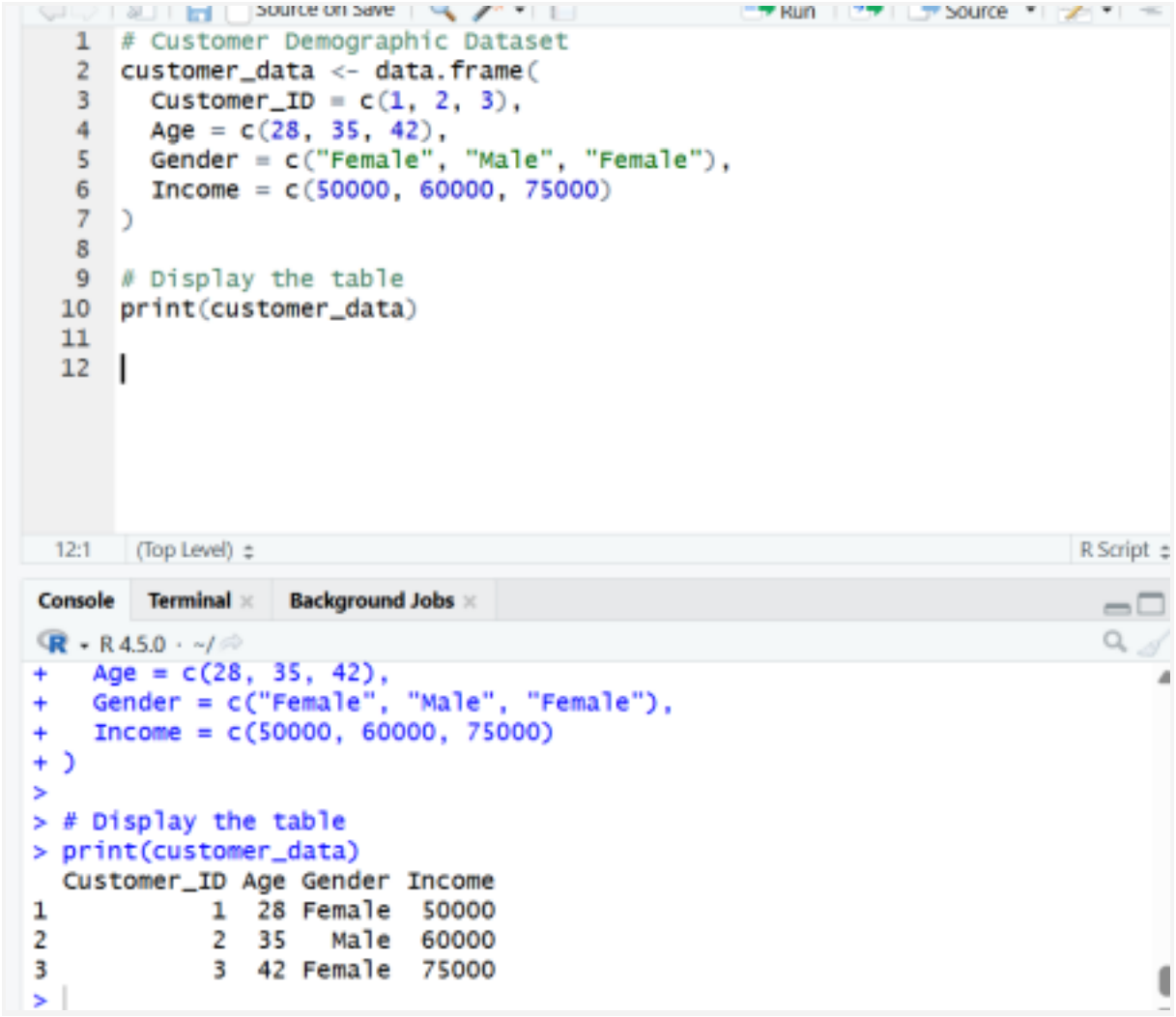
the given dataset.

PROGRAM 3: Customer Demographic Information Table

Question: Build a table to show the demographic information for each customer. Label the table elements.

```
# Customer Demographic Dataset
customer_data <- data.frame(
 Customer_ID = c(1, 2, 3),
 Age = c(28, 35, 42),
 Gender = c("Female", "Male", "Female"),
 Income = c(50000, 60000, 75000)
)

# Display the table
print(customer_data)
```



Output 3: Customer Demographic Information Table

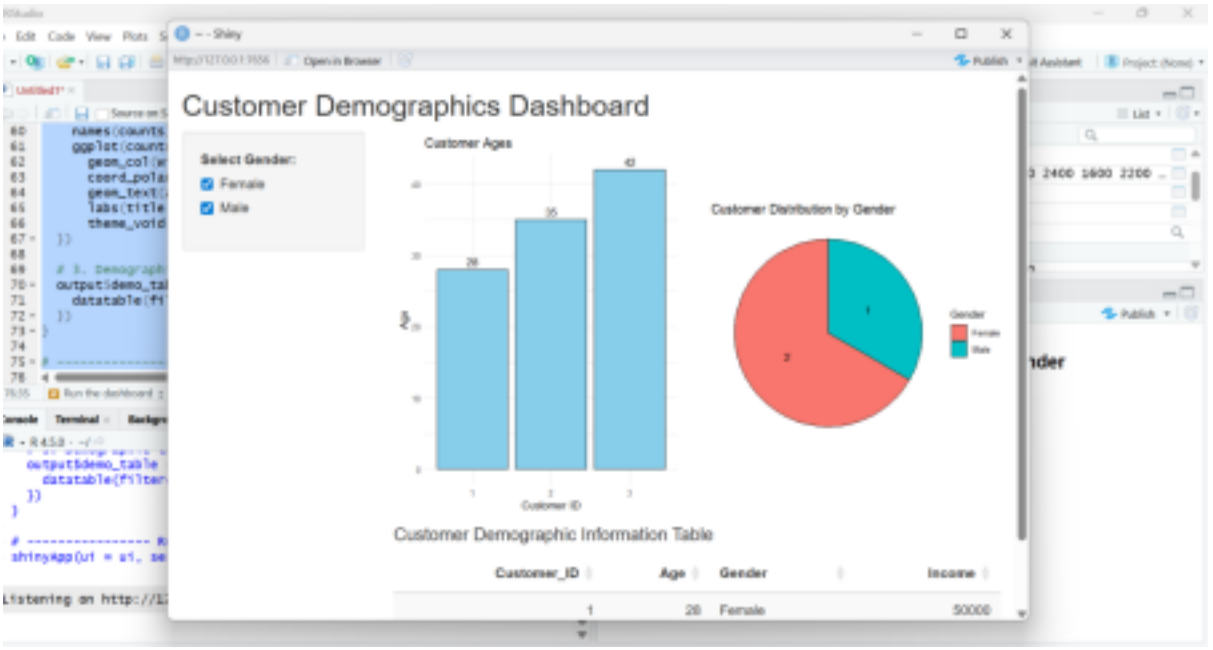
Observation: The table displays the Customer ID, Age, Gender, and Income for each of the three customers.

PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Tableau Steps

1. Open Tableau and connect to the Customer Demographics dataset.
2. Verify the fields Customer ID, Age, Gender, and Income.
3. Create a new worksheet for the Customer Ages bar chart.
4. Drag Customer ID to Columns and Age to Rows, then select Bar in the Marks card.
5. Add the title “Customer Ages” and label the axes.
6. Create another worksheet for the gender pie chart.
7. Drag Gender to the view and use Number of Records (or COUNT(Customer ID)) as the measure; select Pie from the Marks card.
8. Create a third worksheet and drag Customer ID, Age, Gender, and Income into the view to form the demographic table.
9. Create a new Dashboard and add the bar chart, pie chart, and table.
10. Add the title “Customer Demographics Dashboard”.
11. Use the pie chart or a suitable field as a filter where appropriate to make the dashboard interactive.
12. Test the dashboard by selecting a gender/category and checking the linked views.



Output 4: Customer Demographics Dashboard

Observation: The Tableau dashboard combines the customer age bar chart, gender pie chart, and demographic table, allowing interactive exploration of the supplied customer information.

Procedure

1. Create the customer demographic data in R.
2. Prepare the customer age values and create the bar chart using barplot(). 3. Prepare the

gender data, calculate frequencies using table(), and create the pie chart using pie().

4. Create a data frame containing Customer ID, Age, Gender, and Income and display it as a table.
5. Open Tableau and import the Customer Demographics dataset.
6. Create the customer age bar chart and gender pie chart in Tableau.
7. Create the demographic information table in Tableau.
8. Combine the three visualizations into an interactive dashboard and add a suitable title. 9. Test the dashboard using the available filter/interactivity.

Result

The Customer Demographics data was successfully visualized using a bar chart, pie chart, demographic information table, and an interactive Tableau dashboard.